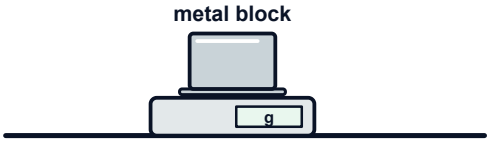
DO NOW, REQUIRED PRACTICAL

Specific heat capacity

Energy
Paper 1, Combined and Separate
6 steps

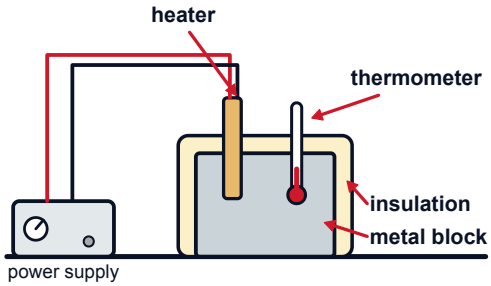
A student measures the specific heat capacity of a metal block.

1 Measure the mass of the block with a balance.

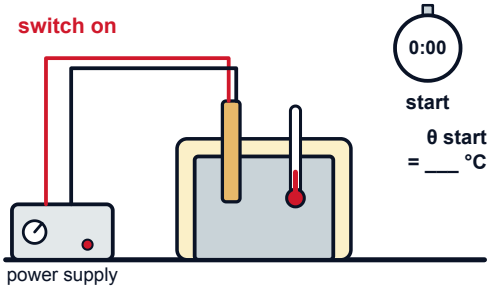


measure its mass m

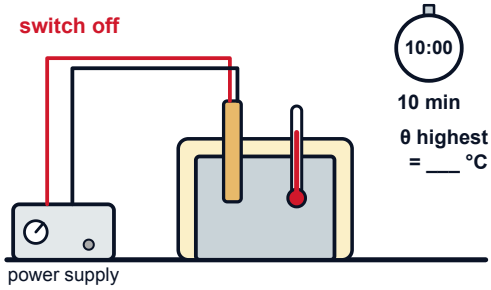
2 Put a heater and a thermometer into the block and wrap it in insulation.



3 Record the starting temperature, then switch on the heater and start a timer.



4 After 10 minutes, switch off and record the highest temperature reached.



5 Calculate the energy supplied from the power of the heater and the time.

$$E = P \times t$$

energy (J) = power (W) × time (s)

time in seconds:
10 minutes = 600 s

6 Calculate the specific heat capacity: energy ÷ (mass × temperature rise).

$$c = E \div (m \times \Delta\theta)$$

J/kg°C = J ÷ (kg × °C)

Δθ = θ highest − θ start

m in kg (not g)

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